MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPES 7YP1, 7YP2, AND 7YP7

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Electrostatic deflection and focus, square face, five beam

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Eb1	Eb2	Eb3	ed	Rg	Zd	Ehk	Eb3/Eb2
Unit:	V	Vdc	Vdc	Vdc	Vdc	v	Meg	Meg	Vdc	Ratio
Maximum:	6.9	-200	1,650	3,850	7,700	1,250	1.5	1.0 (see note 1)	±180	2.0 (see note 2)
Minimum:	5.7	0	---	---	---	---	---	---	---	---
TEST CONDITIONS: (see note 3)	6.3	Adjust	Focus	2,000	4,000	---	---	1.0	---	2.0

GENERAL:

Qualification - Required

Ⓖ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1						
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (electrostatic types)	-		---	---	---	---
5206	Gas "cross"	6,11	Ib3 = 50 µAdc	---	---	---	---
5106	Bulb, screen, and faceplate quality	-		---	---	---	---
5221	Light output	11	P1 screen; Ib3 = 50 µA	Light	25	---	ftL
5223	Modulation	-	Ib3 = 50 µAdc	ΔEc1	---	55	Vdc
5231	Spot position (electrostatic deflection)	8		---	---	16	mm
5231	Spot displacement (leakage)	-		---	---	10	mm
5241	Grid cutoff voltage	-		Ec0	-90	-50	Vdc
5251	Grid No. 1 leakage current	-		Ic1	---	3	µAdc
5251	Anode No. 2 leakage current	-		Ib2	---	10	µAdc
	Quality conformance inspection, part 2						
5221	Screens						
	Type P2	11		D1(1)	360	---	cB
	Type P7	11		D1(5)	400	---	cB
	Type P7	11		G5:1	4	---	cB
1301	Heater current	-		Ib	540	660	mA
5201	Electrode current (anode No. 1)	-		Ib1	-15	+10	µAdc
5201	Electrode current (cathode)	-	Ib3 = 50 µAdc	Ik	---	1,000	µAdc
5101	Base alinement (electrostatic types)	-	3D4; base key	---	---	---	---
5101	Trace and bulb alinement (rectangular electrostatic types)	-	1D2 trace and bulb wall with contact button	---	---	3	Degrees
5101	Side terminal alinement (electrostatic types)	-	3D4	---	---	---	---
5101	Side terminal and base alinement	-	Base key; collar index pin	---	---	---	---
5101	Neck and base alinement (electrostatic types)	-		---	---	---	---
5226	Line width "A" (electrostatic deflection)	-	Ib3 = 25 µAdc	Width	---	0.65	mm
5226	Line width "B" (electrostatic deflection)	7	Ib3 = 25 µAdc	Width	---	0.80	mm
5246	Focusing voltage (zero-bias)	-		Eb1	450	---	Vdc
5246	Focusing voltage at cutoff	-		Eb1	---	650	Vdc
5248	Deflection factor	9	1D2	DF	68	84	Vdc/in.

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 -Continued						
5248	Deflection factor	10	3D4	DF	27	37	Vdc/in.
5251	Heater-cathode leakage current	-		Ihk	---	15	µAdc
5101	Angle between traces	-		---	88	92	Degrees
(G) ---	Trace alinement	12		---	---	---	---
1111	Base pin solder depth	-		---	---	---	---
(G) 1101	Secureness of base, cap, or insert	-		---	---	---	---
(G) 1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	-	Group C; t = 500 hours; Ib3 = 25 µAdc	---	---	---	---
---	Life-test end points:						
5223	Modulation	-	Ib3 = 40 µAdc	ΔEc1	---	55	Vdc
5226	Line width "A"	-	Ib3 = 18 µAdc	Width	---	0.65	mm
5226	Line width "B"	-	Ib3 = 18 µAdc	Width	---	0.80	mm
1141	Pressure (implosion)	5		---	---	---	---
5111	Vibration	5		Width	---	1.0	mm
1331	Direct-interelectrode capacitance	4,5	k to all	Ck	---	6.5	pF
			g1 to all	Cg1	---	7.0	pF
			D1 to D2	C1D2	---	2.7	pF
			D3 to D4	C3D4	---	3.2	pF
			D1 to all	CD1	---	8.1	pF
			D2 to all	CD2	---	8.0	pF
			D3 to all	CD3	---	7.8	pF
			D4 to all	CD4	---	7.8	pF
5101	Neck and bulb alinement (electrostatic types)	5		---	---	---	---
5216	Stray light emission (conventional types)	5	Eb2 = 3,850 Vdc; Eb3 = 7,700 Vdc	---	---	---	---
5216	Cathode illumination	5		---	---	---	---
5248	Deflection-factor uniformity	5	1D2	---	---	2	%
			3D4	---	---	5	%
525C	Interaction factor	5		---	---	10×10^{-6}	in./Vdc
(G) 5115	Shock	5		---	---	---	---
(G) 1216	Base material insulating quality	5		---	---	---	---

NOTES:

1. The deflection electrode circuit resistance should be 1 Megohm or less and approximately equal, otherwise beam shift at high drives can be expected. Higher resistance values up to 5 Megohms may be used for low-beam current operation.
2. This tube is designed for optimum performance when operating at an Eb3/Eb2 ratio of 2.0. Operation at other ratios of Eb3/Eb2 may result in changes in deflection uniformity and pattern distortion.
3. All tests except direct-interelectrode capacitance, vibration, and interaction factor, shall be made on each unit separately. Each deflection plate shall be connected to A2 through a resistance of 1 Megohm.

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NOTES: -Continued

4. All other electrodes contained within the envelope are to be tied together.

(G) 5. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

6. The pattern size shall be adjusted to 1.5 inches in the limited scan direction.

7. Due to limited scan of 3D4 plates, the length of scan in the 3D4 direction shall be limited to ±1 inch from the undeflected spot position. Line width "B" shall be measured at ±0.75 inch from the undeflected spot position.

8. From the spot positions as indicated on figure 1.

9. The deflection in this direction covers the minimum useful screen width following the bulb contour.

10. The deflection in this direction is limited to ±0.75-inch minimum from the undeflected spot position.

11. This test to be performed at the conclusion of the holding period.

(G) 12. Corresponding traces of each unit are to be within 2 degrees of each other.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

User activities:
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Navy - AS, OS, MC, CG
Air Force - 11

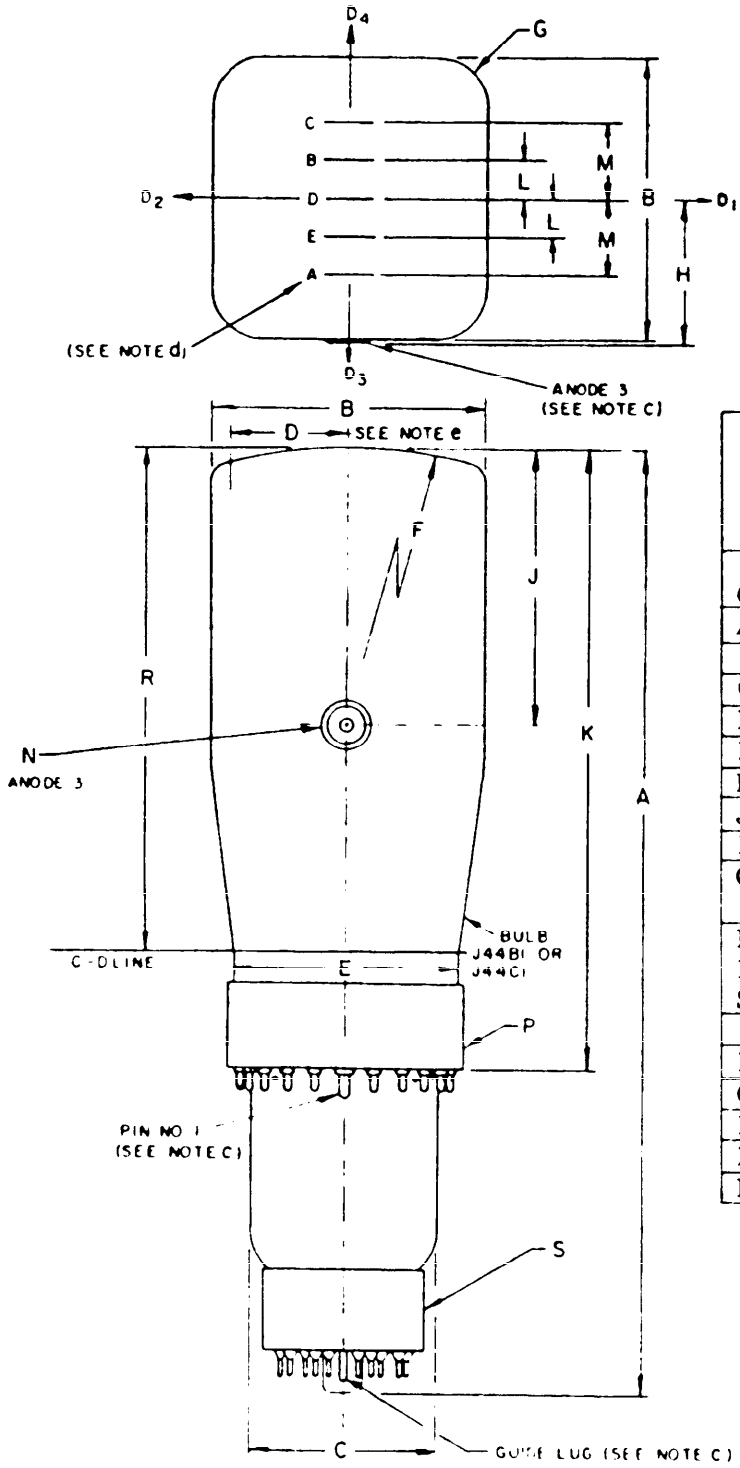
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Agent:
DSA - ES

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not exceed the maximum specified

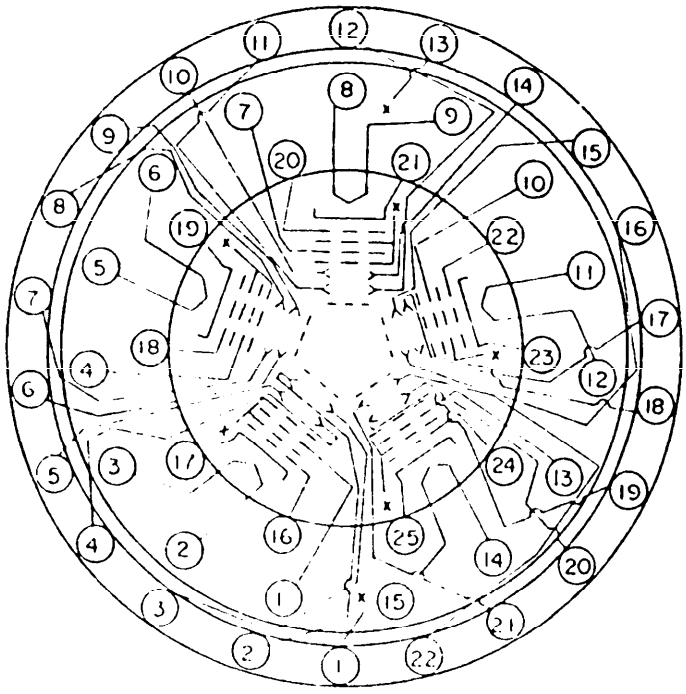


Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	18.125 (460.38)	18.875 (479.43)
B	5.44 (139.18)	5.53 (140.46)
C	3.688 (93.68)	3.813 (96.85)
D	2.25 (57.15)	
E	4.313 (109.55)	4.500 (114.30)
H		2.797 (81.04)
J	5.000 (127.00)	5.50 (139.70)
K	12.125 (307.98)	12.375 (313.33)
Quality conformance inspection, part 3 (See note 5)		
N	Bulb contact: J1-22	
P	Base, collar: A22-101	
S	Base, B25-139 (See note b)	
Reference dimensions (See note a)		
F	40.00 (101.60)	
G	1.00 (25.40)	
L	.75 (19.05)	
M	1.50 (38.10)	
R	10.00 (254.00)	

NOTES:
a. These dimensions are for information and are not required for inspection purposes.
b. For pin alinement use JEDEC gage GB25-1.
c. Bulb contact J1-22, pin number 1 of base A22-101, and guide lug of base B25-139 shall each aline with 3D4 trace within 10°.
d. Beam A shall center on position A as shown within .312 R. This requirement shall also apply to beams B, C, D, and E and their corresponding positions.
e. Useful screen radius.

FIGURE 1. Outline drawing of electron tube types 7YP1, 7YP2, and 7YP7.

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BOTTOM VIEW

22 PIN [illegible] BASE

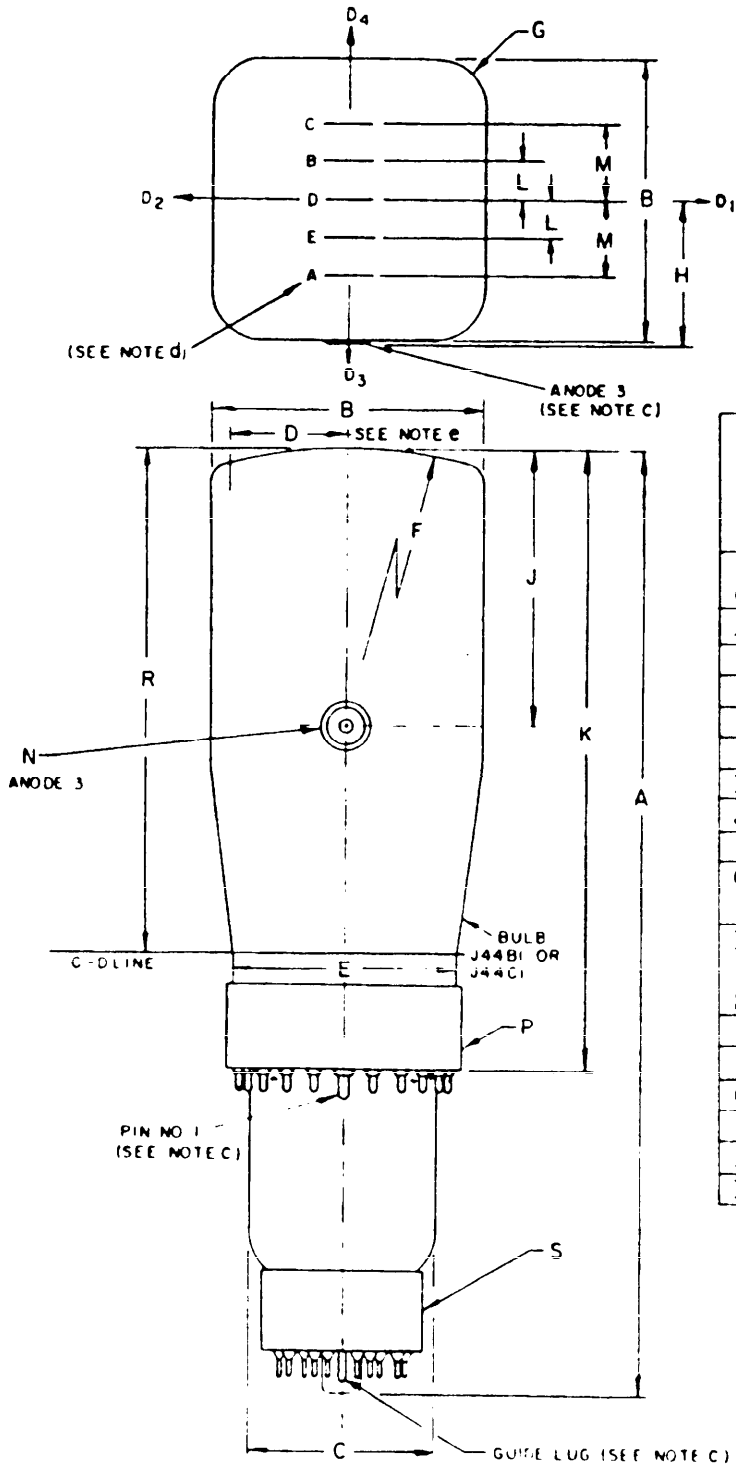
PIN NO	ELEMENT	BEAM
1	ACCELERATOR	
2	DEFLECTOR D2	A
3	DEFLECTOR D	A
4	DEFLECTOR D1	A
5	DEFLECTOR D4	A
6	DEFLECTOR D1	B
7	DEFLECTOR D3	B
8	DEFLECTOR D4	B
9	DEFLECTOR D2	B
10	DEFLECTOR D3	C
11	DEFLECTOR D1	C
12	DEFLECTOR D2	C
13	ACCELERATOR	
14	DEFLECTOR D4	C
15	DEFLECTOR D1	D
16	DEFLECTOR D2	D
17	DEFLECTOR D4	D
18	DEFLECTOR D3	D
19	DEFLECTOR D4	E
20	DEFLECTOR D3	E
21	DEFLECTOR D2	E
22	DEFLECTOR [illegible]	E

25 PIN BASE

PIN NO	ELEMENT	BEAM
1	FOCUSING ELECTRODE	A
2	HEATER	A
3	HEATER	A
4	FOCUSING ELECTRODE	B
5	HEATER	B
6	HEATER	B
7	FOCUSING ELECTRODE	C
8	HEATER	C
9	HEATER	C
10	FOCUSING ELECTRODE	D
11	HEATER	D
12	HEATER	D
13	FOCUSING ELECTRODE	E
14	HEATER	E
15	HEATER	E
16	GRID NO1	A
17	CATHODE	A
18	GRID NO1	B
19	CATHODE	B
20	GRID NO1	C
21	CATHODE	C
22	GRID NO1	D
23	CATHODE	D
24	GRID NO1	E
25	CATHODE	E

FIGURE 1. Outline drawing of electron tube types 7YP1, 7YP2, and 7YP7 - Continued.

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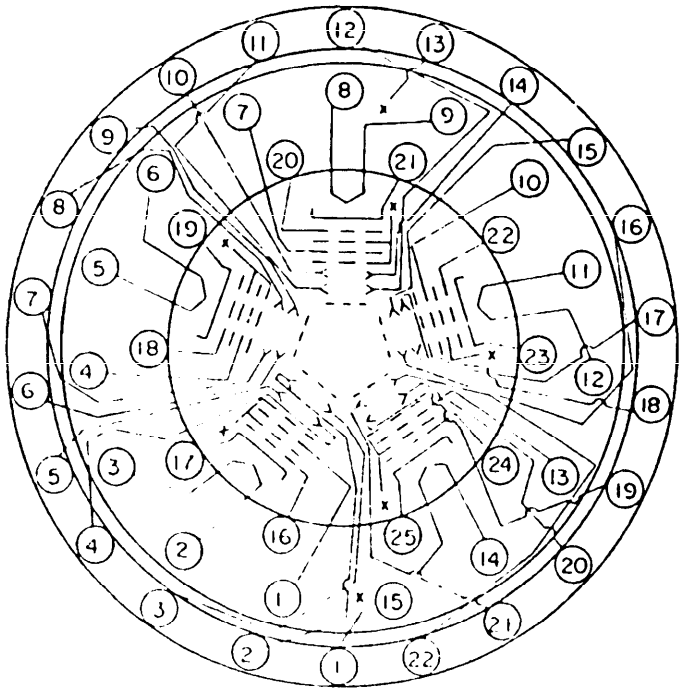


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BOTTOM VIEW

22 PIN [illegible] BASE

PIN NO	ELEMENT	BEAM
1	ACCELERATOR	
2	DEFLECTOR D2	A
3	DEFLECTOR D3	A
4	DEFLECTOR D1	A
5	DEFLECTOR D4	A
6	DEFLECTOR D1	B
7	DEFLECTOR D3	B
8	DEFLECTOR D4	B
9	DEFLECTOR D2	B
10	DEFLECTOR D3	C
11	DEFLECTOR D1	C
12	DEFLECTOR D2	C
13	ACCELERATOR	
14	DEFLECTOR D4	C
15	DEFLECTOR D1	D
16	DEFLECTOR D2	D
17	DEFLECTOR D4	D
18	DEFLECTOR D3	D
19	DEFLECTOR D4	E
20	DEFLECTOR D3	E
21	DEFLECTOR D2	E
22	DEFLECTOR D1	E

25 PIN BASE

PIN NO	ELEMENT	BEAM
1	FOCUSING ELECTRODE	A
2	HEATER	A
3	HEATER	A
4	FOCUSING ELECTRODE	B
5	HEATER	B
6	HEATER	B
7	FOCUSING ELECTRODE	C
8	HEATER	C
9	HEATER	C
10	FOCUSING ELECTRODE	D
11	HEATER	D
12	HEATER	D
13	FOCUSING ELECTRODE	E
14	HEATER	E
15	HEATER	E
16	GRID NO1	A
17	CATHODE	A
18	GRID NO1	B
19	CATHODE	B
20	GRID NO1	C
21	CATHODE	C
22	GRID NO1	D
23	CATHODE	D
24	GRID NO1	E
25	CATHODE	E

FIGURE 1. Outline drawing of electron tube types 7YP1, 7YP2, and 7YP7 - Continued.

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